ĐẠI HỌC TÔN ĐỨC THẮNG TON DUC THANG UNIVERSITY	KHOA CÔNG NGHỆ THÔNG TIN KỲ THI GIỮA KỲ HỌC KỲ 1 NĂM HỌC: 2026 – 2027

ESSAY: Introduction to Digital Image Processing

CODE: 505060

I. Rules

- Only use **OpenCV, math, and Numpy** libraries in the Programming part.
- You are **not allowed to use pre-trained models** in the Programming part.
- Students are strictly **prohibited from hardcoding solutions** or writing specialized logic tailored to specific video frames.
- **Use the same source code** for all input videos.
- Each output video must have **same color model, frame size, and frame rate** with the corresponding input video.
- Output videos must be in **MP4** format.
- Violated any above rule will be scored 0.

II. Programming part (7.0 points)

Given two input videos with filenames: **video1.mp4** and **video2.mp4**

The input video files are in the following link:
https://drive.google.com/drive/folders/15IMKKVqdRqsybV7-hVhwEgIMch1mKz1H?usp=sharing

Draw a rectangular bounding box surrounding each **circular** Traffic sign in all frames of the input videos **automatically** (only traffic signs **in the centre or right side of the street**). Finally, save the outputs into new video files.

- Input: two input videos (video1.mp4 and video2.mp4)
- Output: another version of the input videos with a rectangle surrounding each circular traffic sign. A sample output was attached in the assignment ("*video1_output_sample.mp4*").

You can see a guidline here:
https://docs.google.com/document/d/1MqhsAVoE6vnGLOpFDv4zpBadQahU-XrcW4O9BUZ7SpI/edit?usp=sharing

III. Report part (3.0 points)

1. The report must be submitted in **PDF format**, and the content must be written based on the report/essay format of the Faculty of Information Technology. **In case students do not follow the Faculty's format, they will receive 0 points for the Report part.**
2. The report must include the following contents:
    a. Chapter 1: Methodology of Solving Tasks **(2.5 points)**

    Write a detailed description of the solving methods used in the "Programming part" by:

    - Providing a detailed explanation of main programming steps in the task
        - In each step, provide the corresponding source code and its explanation, and an example of output images/information

    b. Chapter 2: Task results **(0.5 point)**

    - For each input video, insert 05 output frames which have circular traffic signs in 05 different time of the corresponding output video (minimum difference of time between two consecutive output frames is 3 seconds)
    - Insert the sharing link of the output videos

    The images must be clear, and properly laid out. The images captions and descriptions are also required.

IV. Submission guideline

- Filenames of the source code and the report files must be the **Student IDs**, for ex.,
    - A group of only one student with student ID 521H1495 will submit a Python source file named **521H1495.py** and a report file named **521H1495.pdf**
    - A group of two students with student IDs 521H1234 and 522H4321 will submit a Python source file named **521H1234_522H4321.py** and a report file named **521H1234_522H4321.pdf**
- Filenames of the output videos must include the **Student IDs**, and **insert Student IDs** into all frames of the output videos.
- Upload output videos to Google Drive and **share them publicly** (anyone on the internet with the link can view them). In the Chapter 2 of report, you insert the link of the output videos.
- Students submit **Python source files, and a report file** to the "**MidTerm_Essay**" assignment on Elearning website of the practical class.
- Python source files must be saved in the correct format (**file extension is .py**).
- **All members must submit the essay.**

V. Regulations

- The result of this essay will be the Midterm score.
- Student groups that submit their projects **late** will receive 0.0 points for each group member.
- **Missing any required materials** from the submission will receive 0.0 points for each group member.
- **NOT inserting the sharing link of the output videos or cannot access them** will receive 0.0 points for each group member.
- Students must ensure that the Python source files are not corrupted during execution. **The source code with errors or in the wrong format** will receive 0.0 points for each group member.
- **Copying** source code from the Internet or other students, sharing your work with other groups, or engaging in similar forms of academic misconduct will result in a score of 0.0 for all groups involved.
- If a student's work shows signs of copying each other, the student will attend an interview with the lecturer.

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